

Notes: Seru (JFE, 2014)
Firm Boundaries Matter: Evidence from Conglomerates and R&D Activity

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1 Big picture

- **Question.** Do firm boundaries affect the allocation and productivity of innovative activity? The paper asks whether becoming part of a diversified conglomerate changes the scale and novelty of a target firm’s R&D output.
- **Core mechanism.** Conglomerate headquarters can reallocate capital across divisions. This “visible hand” may improve hard-information capital budgeting, but it can weaken incentives for novel R&D because researchers and divisional managers fear that resources will be diverted after interim signals arrive.
- **Main prediction.** The negative effect should be strongest when the acquirer operates an active internal capital market (ICM), because the threat of reallocating resources is more credible and more salient.
- **Empirical setting.** The paper uses Compustat segment data, NBER/USPTO patent and citation data, and SDC merger data over 1980–1998. Innovation is measured using patents and citations per patent, adjusted for technology-class and cohort effects.
- **Suggestive aggregate evidence.** Among multi-segment firms, greater ICM intensity is associated with less-novel innovation. Measures such as R&D competition among divisions, segment diversity, and capital reallocation are negatively related to citations per patent.
- **Identification strategy.** The core design uses failed friendly mergers as a control group for successful mergers. Failed targets provide a counterfactual for what successful targets would have looked like had they not entered the acquirer.
- **Main causal result.** Relative to failed targets, targets acquired in successful mergers suffer a large post-merger decline in R&D productivity. The decline is concentrated in diversifying/conglomerating mergers, not in non-conglomerating mergers.
- **ICM heterogeneity.** The treatment effect is larger when the acquiring conglomerate had a more active internal capital market before the merger. This directly ties the R&D decline to the internal resource-allocation process.
- **Inventor margin.** The decline is not mainly due to inventors leaving. Inventors are less productive after being absorbed into diversified firms, so the key margin is intensive productivity rather than only extensive inventor exit.
- **Organizational response.** Acquirers partly mitigate the problem by moving innovative activity outside the firm boundary through strategic alliances and joint ventures. Additional evidence from IRI survey data suggests that decentralized R&D budgets are associated with more novel R&D inside conglomerates.
- **Economic takeaway.** Firm boundaries matter because internal capital allocation changes incentives. Conglomerates may be useful for some forms of investment, but active internal capital markets can suppress novel research inside the firm.

2 Incremental contribution of the paper

- **From conglomerate value to internal resource allocation.** Earlier conglomerate work often compares market values or productivity across stand-alone and diversified firms. This

paper studies a specific internal activity - innovation - and asks whether organizational form causally affects R&D productivity.

- **A causal design for firm boundaries.** The failed-merger quasi-experiment moves beyond cross-sectional comparisons of conglomerates and stand-alone firms. It uses acquisition outcomes that are plausibly unrelated to target R&D quality to difference out selection into conglomeration.
- **A direct test of internal capital market theories.** The paper distinguishes the bright-side view of ICMs from the dark-side view. It shows that the firms most likely to use active ICMs are also those in which target innovation declines most after acquisition.
- **A focus on novelty, not only R&D spending.** The paper does not rely on R&D expenditures alone. It uses patent counts and citation-weighted novelty, which better capture research output and impact.
- **Inventor-level mechanism.** By hand-collecting inventor information, the paper decomposes the treatment effect into inventor exit and within-inventor productivity. This helps identify how organizational form affects research: inventors who remain become less productive.
- **Boundary response.** The paper shows that acquirers respond by using alliances and joint ventures, suggesting that conglomerates may shift uncertain R&D outside their own hierarchy when internal incentives are weak.
- **Organizational design within conglomerates.** The IRI survey evidence links R&D budget decentralization to more novel innovation, connecting the merger evidence to a concrete managerial design choice.
- **Broader contribution.** The paper provides empirical evidence that the allocation of control rights and capital inside the firm shapes the type of knowledge production that occurs inside versus outside firm boundaries.

3 Data and measurement

3.1 Firm and segment data

- The base firm sample comes from Compustat's industry segment files for 1980–1998.
- The paper excludes financial and government segments, firms with missing segment information, very small firms, and observations that fail standard Berger-Ofek segment aggregation screens.
- The resulting sample contains 12,090 diversified firm-years and 32,018 single-segment firm-years.
- Segment information is used to characterize conglomerates and to construct measures of internal capital market intensity.

Interpretation: The segment data identify whether a firm is diversified and provide the building blocks for measuring how much headquarters might reallocate resources across divisions.

3.2 Measures of ICM intensity

The paper uses several proxies for how important internal capital markets are inside a conglomerate.

- **Reallocate.** Measures the gap between cash-surplus and cash-deficit divisions. A higher value means internal cross-subsidization opportunities are larger.
- **Diversity.** Measures dispersion in segment investment opportunities, following Rajan, Servaes, and Zingales. Higher dispersion implies a stronger role for internal capital allocation.
- **Diversification index.** The inverse Herfindahl index of segment sales. Higher values indicate more fragmented conglomerates.
- **Compete.** Measures competition for R&D resources across divisions using the inverse Herfindahl index of prior-year patenting by divisions.

Interpretation: The important empirical idea is that not all conglomerates are the same. The dark-side prediction is not simply about being diversified; it is about being diversified in a way that makes active internal reallocation likely.

3.3 Innovation and patent data

- Innovation measures come from the NBER patent data set, based on Hall, Jaffe, and Trajtenberg, augmented with detailed USPTO information.
- The paper uses 423,640 patents granted during the sample period.
- **Patent** is the adjusted number of patents. It measures the scale of innovative output.
- **CPatent** is adjusted citations per patent. It measures the novelty or impact of innovations.
- Adjustments correct for truncation in patent grants and citations, as well as technology-class and cohort differences.
- To attribute patents to divisions, the paper uses assignee names, inventor locations, subsidiary locations, technology classes, and other patent-level details.

Interpretation: Forward citations are used as a proxy for patent importance. A decline in citations per patent means the firm is not merely producing fewer patents; it is producing less-novel or less-influential research.

3.4 Quasi-experiment sample

- The treatment group consists of firms acquired in successful friendly mergers from SDC Platinum.
- The control group consists of targets in friendly mergers that failed for reasons unrelated to the R&D activity of the target or bidder.
- The paper reads news articles from Lexis-Nexis and Factiva to classify why a deal failed.
- Deals are retained in the control group when failure is due to reasons such as regulatory objections, competing offers, unexpected legal events, market conditions, or exogenous events such as the 1987 crash.

- The final analysis uses 510 treatment firms and 170 control firms, with years before and after the merger event.

Interpretation: Failed mergers are useful because their targets were selected into merger attempts, but they did not actually enter the acquiring organization. This helps separate the effect of being acquired from the effect of being the kind of firm that attracts acquirers.

3.5 Inventor, alliance, and R&D budget data

- Inventor-level tests use hand-collected information on inventors responsible for patents in the quasi-experiment sample.
- Strategic alliances and joint ventures are collected from SDC and are used to test whether R&D activity is moved outside the acquirer after merger.
- R&D budget decentralization comes from Industrial Research Institute surveys in 1994 and 2001.
- The IRI data identify whether R&D is divisional or centralized and what share of R&D budget is controlled by headquarters.

Interpretation: These additional data sets turn the paper from a reduced-form acquisition study into a mechanism paper: they show who becomes less productive and how firms adjust their organizational boundaries.

4 Models and empirical specifications

4.1 Aggregate ICM intensity regression

In the aggregate sample of multi-segment firms, the paper estimates specifications of the form

$$y_{it} = \alpha + \beta_1 ICMIntensity_{it} + \delta Z_{it} + \mu_t + \mu_j + \varepsilon_{it},$$

where y_{it} is either $CPatent$ or segment-industry-adjusted $CPatent$, Z_{it} includes firm controls, μ_t are year effects, and μ_j are industry effects.

- The hypothesis is $\beta_1 < 0$: more active internal capital markets should be associated with less-novel R&D.
- Controls include firm size, R&D, leverage, cash, profitability, capital expenditures, asset tangibility, Tobin's Q, firm age, and industry concentration.
- This analysis is suggestive because the organizational form is not randomly assigned.

Interpretation: Table 2 is the first evidence that ICM intensity and innovation novelty are negatively related, but the paper reserves causal interpretation for the failed-merger design.

4.2 Selection problem and treatment effect

The paper frames the causal problem using the average treatment effect:

$$ATE = E[y_i(C = 1) - y_i(C = 0)],$$

where $C = 1$ means the firm is part of a conglomerate and $C = 0$ means it is not.

A simple comparison of conglomerate and stand-alone firms contains selection bias:

$$E[y_i(1)|C = 1] - E[y_i(0)|C = 0] = \underbrace{E[y_i(1)|C = 1] - E[y_i(0)|C = 1]}_{\text{treatment effect on treated}} + \underbrace{E[y_i(0)|C = 1] - E[y_i(0)|C = 0]}_{\text{selection bias}}.$$

- Firms that become part of conglomerates may differ from firms that remain stand-alone.
- Acquirers may select targets with declining, maturing, or unusually promising R&D pipelines.
- Failed mergers create a control group that helps remove this selection component.

Interpretation: The central empirical challenge is not whether acquired targets change after the merger, but whether they change relative to comparable targets that almost merged but did not.

4.3 Validation of failed-merger control group

The paper estimates a logit model:

$$Pr(Success_{it} = 1) = \Phi(\alpha + \gamma X_{i,t-1} + \beta_1 R\&DProductivity_{i,t-1} + \mu_t).$$

- *Success* equals one if the deal succeeds and zero if it fails.
- The key coefficient is on lagged R&D productivity.
- The paper finds that pre-merger patents and citations do not predict deal success conditional on observables.

Interpretation: This supports the assumption that failed deals are not systematically failed because their targets had different R&D productivity.

4.4 Main difference-in-differences specification

The core specification is

$$CPatent_{it} = \alpha + \gamma_1 After_{it} + \gamma_2 (After_{it} \times T_i) + \delta Z_{it} + \mu_t + \mu_i + \varepsilon_{it},$$

where *After* equals one after the merger event and T_i equals one for successful mergers.

- γ_1 captures the post-event change for failed targets.
- γ_2 captures the extra post-event change for successfully acquired targets.
- A negative γ_2 means successful acquisition reduces target R&D productivity relative to failed targets.

Interpretation: This is the paper's main causal specification. Firm fixed effects remove time-invariant target quality, and the failed-merger group controls for normal post-attempt dynamics.

4.5 Heterogeneity by merger type and ICM intensity

The paper extends the DD model by interacting $After \times T$ with indicators for unrelated mergers, conglomerating acquirers, and acquirer ICM intensity:

$$CPatent_{it} = \alpha + \gamma_1 After_{it} + \sum_k \gamma_k (After_{it} \times T_i \times H_{ik}) + \delta Z_{it} + \mu_t + \mu_i + \varepsilon_{it}.$$

- H_{ik} includes unrelated-merger status, whether the acquirer is multi-segment, and measures such as *Compete*, *Diversity*, *Diversification*, and *Reallocate*.
- The prediction is that the treatment effect is concentrated in diversifying mergers and is larger when acquirer ICM intensity is high.

Interpretation: These tests connect the treatment effect to the theory. If the effect is really about firm boundaries and internal capital allocation, it should be strongest in mergers that bring the target inside an active conglomerate ICM.

4.6 Inventor-level and mitigation specifications

Inventor-level tests use outcomes such as whether an inventor remains present after the merger and the average $CPatent$ of an inventor's patents. The basic structure is again a difference-in-differences design with inventor controls.

Mitigation tests use the same event-study logic with outcomes such as combined market value, ROA, cash, R&D-to-sales, and indicators for whether the acquirer enters strategic alliances or joint ventures after the merger.

Interpretation: These specifications ask two mechanism questions: does the decline come from inventor exit or lower productivity, and do acquirers compensate by changing organizational form?

4.7 R&D budget decentralization

The IRI survey analysis estimates

$$CPatent_{it} = \alpha + \beta_1 \%HqBudget_{it} + \beta_2 Divisional_{it} + \beta_3 (\%HqBudget_{it} \times Divisional_{it}) + \delta Z_{it} + \mu_t + \varepsilon_{it}.$$

- $\%HqBudget$ is the share of R&D budget financed or controlled by headquarters.
- *Divisional* indicates whether the firm conducts R&D at the divisional level.
- The key prediction is that more headquarters control should be associated with less-novel R&D, especially when R&D is divisional.

Interpretation: If the central problem is headquarters' inability to commit not to reallocate resources, then decentralized R&D budgets are a natural organizational response.

5 Identification and empirical design

5.1 What the design can identify

- The failed-merger design identifies the effect of successful acquisition relative to the counterfactual path of similar targets whose mergers failed.
- The strongest causal interpretation applies to innovative firms that were merger targets and to cases where merger failure is plausibly unrelated to R&D quality.
- The heterogeneity tests identify whether the post-merger effect is concentrated in situations where theory predicts greater ICM agency problems.

Interpretation: The design is not a random assignment of all firms to conglomerates. It is a quasi-experiment within the population of merger targets.

5.2 Why failed mergers are useful

- Both successful and failed targets were attractive enough to receive merger bids.
- Failed targets remain outside the acquiring organization, so their post-event R&D provides a counterfactual for acquired targets.
- The paper screens failed deals to exclude cases where R&D itself caused the failure.
- The validation tests show that lagged patents and citations do not predict deal success.

Interpretation: The key assumption is that, conditional on observables, the reason a target ends up in the treatment rather than control group is not driven by future R&D productivity.

5.3 Remaining threats and limitations

- Some unobservable difference between successful and failed deals could still remain despite the screening process.
- Patent citations are noisy measures of innovation quality and may vary by industry, technology, or time despite adjustments.
- The inventor matching process is complex and may contain some measurement error.
- The estimates apply most directly to merger targets with patenting activity, not necessarily to all conglomerate divisions or all types of investment.
- The paper shows little change in overall firm value, so it cannot claim that conglomeration is always value destroying; the result is about the location and novelty of R&D output.

Interpretation: The paper is strongest as evidence that internal capital allocation can impair novel R&D. It is not a blanket argument that all conglomerates are inefficient.

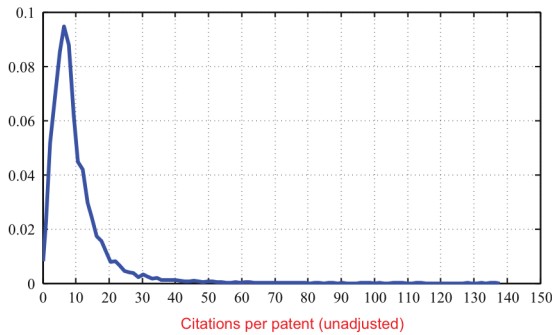
6 Tables and figures

6.1 Figure 1 and Table 1: Innovation distribution and summary statistics

Read:

- Figure 1 shows that citations per patent are highly skewed. Most patents receive few citations, while a small right tail receives many.
- This skewness motivates using citations per patent as a measure of novelty or economic importance rather than treating all patents equally.
- Table 1 shows that most firm-years produce zero patents; patenting is concentrated in a small subset of firms.
- Multi-segment firms are larger and spend more on R&D, but their patents receive fewer adjusted citations per patent than single-segment firms.
- Among patenting multi-segment firms, those with below-mean *CPatent* have higher ICM-intensity proxies such as *Compete*, *Diversity*, and *Reallocate*.

Economic message: Innovation quality is highly uneven, and simple patent counts are not enough. The descriptive evidence already suggests a negative association between active internal capital markets and novel innovation.



(a) Figure 1: Citations per patent are extremely right-skewed.

Table 1
Summary statistics

This table reports the summary statistics of the key variables used in the analysis. Patent information comes from the NBER patent data set, significantly augmented with patent-specific details from the United States Patent and Trademark Office Web site. List of all public firms from the NBER patent file that have financial data available in the S&P's Compustat database and include all the firms in Compustat that operate in the same industries as the firms in the patent database but which do not have patents. Then I apply the same screens as *Boyer and Cox (1995)*. Data on Sales, R&D expenditures, SIC codes, and other financials come from Compustat. Panel A provides details in descriptive statistics of patents across years (firm-years). Panel B corresponds to firm-years for single- and multi-segment firms. Patent and citations per patent measures are adjusted as explained in Appendix A. Among the multi-segment firms that patent, Panel C corresponds to firm-years with above- and below-mean adjusted citations per patent in the sample period. *Signifies that the data rate exceeds firms that produce at least one patent during a given year. Data in this table are for the period 1980-1998.

Panel A: Patenting firm-years over time		Number of patents					Observations
Year		0	1-2	3-10	11-100	> 100	
1980-1985	8,448	1,025	846	659	133	12,011	
1985-1990	8,566	999	906	636	139	11,451	
1990-1995	9,545	963	770	575	106	11,076	
1995-1998	7,940	432	493	440	136	9,416	
All firms	34,838	3,379	3,008	2,349	534	44,108	
Multi-segment firms	7,954	1,250	1,344	1,251	291	12,090	
Single-segment firms	26,884	2,129	1,664	1,098	243	32,018	

Panel B: Single-segment vs. multi-segment firm-years

	Single-segment			Multi-segment			All firms
	Mean (1)	Max (2)	Min (3)	Mean (4)	Max (5)	Min (6)	
Sales (\$ millions)	1,056	17,410	10	3,225	23,512	11	1,595
R&D (\$ millions)	23	239	0	12	878	0	35
Q-Score ^a	1.31	9.35	0.39	0.87	8.49	0.55	1.11
Q-Score ^b	3.54	297	0	1.85	801	0	3.65
CPatent ^c	1.05	5.43	0	0.85	3.45	0	0.96
Observations		32,018			12,090		44,108

Panel C: Patenting multi-segment firm-years

	CPatent > Mean (=0.85)			CPatent < Mean (=0.85)			Patenting firms (7)
	Mean (1)	Max (2)	Min (3)	Mean (4)	Max (5)	Min (6)	
Sales (\$ millions)	3,397	12,836	10	3,774	23,512	19	3,567
R&D (\$ millions)	67	188	0.58	11	878	1.05	96
Compete	2.20	22.86	1.00	2.10	39.70	1.00	2.26
Diversification index	2.44	6.69	1.02	2.40	7.07	1.02	2.43
Diversity	0.85	1.75	0.03	0.78	1.80	0.02	0.80
Reallocate	0.021	0.40	0	0.027	0.62	0	0.039
Observations		2,076			2,069		4,136

(b) Table 1: Summary statistics for patenting and conglomerate variables.

6.2 Table 2 and Table 3: Aggregate ICM evidence and quasi-experiment construction

Read:

- Table 2 relates multi-segment firm R&D productivity to ICM intensity in the aggregate data.
- Compete*, *Diversity*, and *Reallocate* are negatively related to *CPatent* or adjusted *CPatent*, consistent with the prediction that active ICMs reduce R&D novelty.

- The table is only suggestive because diversified firms and their divisions are endogenously selected.
- Table 3 documents how the failed-merger control group is built. The paper starts with unsuccessful bids and excludes failures whose reasons may be related to R&D.
- The final control group contains 175 failed deals after screening.

Economic message: Table 2 motivates the hypothesis; Table 3 creates the empirical design needed to test it more credibly. The paper moves from correlation to quasi-experimental variation.

Table 2
Suggestive evidence: R&D productivity in aggregate data.

This table reports the results of regressions that relate citations per patent produced in a multi-segment firm in a given year to its ICM intensity in that year. The measures of R&D productivity are corrected for truncation in grants and citations as well as for the technology class as explained in Appendix A. The unit of observation is a firm-year. Columns 1-4 use *CPatent* as the dependent variable, while columns 5-8 use segment-industry-adjusted citations per patent (*CPatent^{adj}*) as the dependent variable. ICM intensity is proxied iteratively by *Compete*, *Diversification index*, *Diversity*, and *Reallocate*. *Compete* is only defined for conglomerates that patent at least once during the sample period. Control variables included in the estimation but unsupported for brevity are Sales, RD, Leverage, *EBITDA/Assets*, *Cogs/Assets*, *Cash/Assets*, *NPPE/Assets*, *HE*, *H₂*, *Q*. Data are for the period 1980-1998. ***, **, and * denote significance at 1%, 5%, and 10%, respectively.

	Variation within multi-segment firms							
	<i>CPatent</i>				<i>CPatent^{adj}</i>			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Compete	-0.273*** (0.038)				-0.285*** (0.038)			
Diversification index		-0.008 (0.006)				-0.006 (0.007)		
Diversity			-0.032* (0.016)				-0.041*** (0.017)	
Reallocate				-0.231*** (0.095)				-0.586*** (0.099)
Observations	3,981	12,090	12,090	12,090	3,981	12,090	12,090	12,090
R ²	0.216	0.210	0.234	0.231	0.065	0.107	0.102	0.104
Other controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes	No	No	No	No

(a) Table 2: ICM intensity and R&D productivity in aggregate data.

Table 3
Sample construction for quasi-experiment.

The sample of friendly deals from SDC is selected with the following restrictions: (i) the announcement date falls between 1980 and 1998 and the bidder and target are U.S. public firms; (ii) the bidder's market capitalization exceeds that of firms in the bottom decile using NYSE size breakpoints; and (iii) the mode of payment is all-cash or all-equity. To construct the control group, I start with the sample containing all failed bids and employ the information from Lexis-Nexis and Factiva to exclude any deal whose failure was related to R&D of either the target or the bidder. To focus on deals which are ex ante not too different, I drop deals that the media/analysts/markets expect to fail.

Deals which failed for reasons exogenous to R&D of the target or bidder	
422	All unsuccessful merger bids
-64	Difference in corporate philosophy over growth strategy (not involving R&D)
-45	Other competing bids emerged and the acquisition with the competitor went through
-32	Valuation issues/due diligence revelation about target's operations
-29	Target refusal of offer based on disagreement over valuation
-28	Target and bidder management disagreement over restructuring and strategy (not involving R&D)
-19	Market/analysts expected the deal to fail
-16	Problem in bidder's operations (not involving R&D) revealed over the course of negotiations
-14	Not enough information/negotiations not completed/exogenous events (e.g., 1987 crash)
175	Final control group

(b) Table 3: Construction of failed-merger control group.

6.3 Table 4 and Figure 2: Validating the control group and control-target dynamics

Read:

- Table 4 Panel A compares successful and failed targets before the event. The groups are similar in size, leverage, R&D intensity, citations per patent, and patents.
- Table 4 Panel B shows that lagged patenting and citation-based R&D productivity do not predict whether a deal succeeds.
- This supports the failed-merger control design.
- Figure 2 shows the density of *CPatent* for the control group before and after the intended merger date.
- The control-group distribution is broadly similar before and after the event, with no strong collapse in R&D productivity.

Economic message: The control group appears credible: failed targets do not look systematically different on pre-event R&D, and their post-event innovation distribution does not mechanically deteriorate.

Table 4

Validation of control group construction.

This table reports tests related to validity of control group construction in the quasi-experiment. In Panel A, I provide summary statistics of the control and treatment sample. * Signifies that data are only for firms that produce at least one patent during the sample period before the merger. Panel B presents the logit regression relating the probability of a deal succeeding for a target to the characteristics of the target and the potential acquirer (observation is firm-year). $Dummy^{sameSC}$ and $Dummy^{sameSC}$ take a value one if the potential acquirer is in the same SIC as the target and when the potential acquirer is a multi-segment firm, respectively. HI^{acq} and $CPatent^{acq}$ represent the Herfindahl index and adjusted citations per patent for the potential acquirer. The standard errors in the regressions are robust. Merger dates in the sample correspond to the period 1980-1998. ***, **, and * denote significance at 1%, 5%, and 10%, respectively.

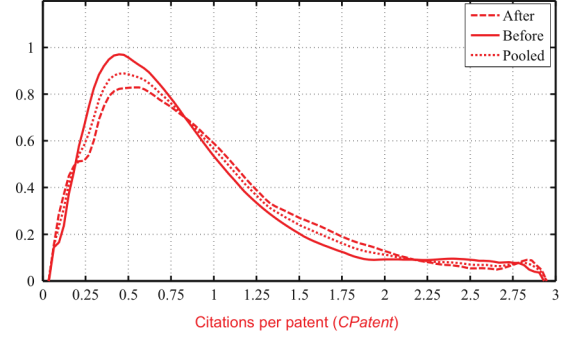
Panel A: Failed and successful targets before the event

	Treatment (successful)	Control (failed)
	Mean (1)	Mean (2)
Size	4.52	4.19
Leverage	0.39	0.38
EBITDA Assets (%)	12.7	10.8
RD Sales (%)	3.5	4.0
CPatents ^a	0.70	0.74
Patents ^a	723	81
Observations	5,200	970

Panel B: Probability of deal succeeding

	Prob(Success = 1)			
	(1)	(2)	(3)	(4)
Size	0.214 (0.151)	0.188 (0.126)	0.118 (0.126)	-0.089 (0.144)
Leverage	-0.278 (0.609)	-0.264 (0.612)	0.0910 (0.601)	0.503 (0.803)
RD	0.008 (0.007)	0.009 (0.007)	0.006 (0.006)	0.009 (0.006)
Patent	-0.001 (0.016)	-0.012 (0.012)	-0.015 (0.012)	-0.014 (0.014)
CPatent	0.475 (0.344)	0.469 (0.343)	0.435 (0.337)	0.595 (0.433)
EBITDA Assets	0.284 (0.303)	0.284 (0.301)	0.281 (0.307)	0.283 (0.306)
HI		1.280 (0.915)	1.060 (0.903)	-0.408 (1.193)
Dummy ^{same}			0.050 (0.300)	0.480 (0.375)
Dummy ^{overlaid}				0.309 (0.183)
HI ^{acq}				1.770 (1.458)
CPatent ^{acq}				0.518 (0.377)
Observations	6,170	6,170	6,170	6,170
Other controls	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes

(a) Table 4: Validation of failed-merger control group.



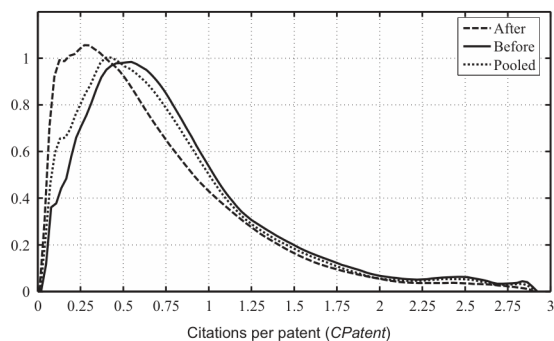
(b) Figure 2: Control group before and after failed merger event.

6.4 Figure 3 and Table 5: Treatment-target dynamics and main DD result

Read:

- Figure 3 shows a clear leftward shift in the treatment group's $CPatent$ distribution after the merger event.
- The shift means acquired targets produce less-novel patents after being absorbed by the acquirer.
- Table 5 provides the formal difference-in-differences test.
- The interaction $After \times T$ is negative and significant for $CPatent$ and for patent counts.
- Columns using acquirer innovation outcomes show little evidence that the target's lost innovation simply reappears inside the acquirer.

Economic message: The main empirical result is that successful acquisition reduces the target's innovative output and novelty relative to failed mergers. The effect is not just a relabeling of R&D from target to acquirer.



(a) Figure 3: Treatment group before and after successful merger.

Table 5
Difference-in-differences: R&D productivity of targets in treatment and control groups.
This table reports the main results from the quasi-experiment. A difference-in-differences specification is employed with R&D productivity measures defined in Appendix A as the dependent variables. The unit of observation is a firm-year. *After* is a dummy variable that takes a value one after the merger event and zero otherwise. *T* is an indicator variable that takes a value one for the treatment group (successful mergers) and zero for the control group (failed mergers). $CPatent^{Treat}$ and $CPatent^{Ctrl}$ represent adjusted citations per patent and patents of the acquirer, respectively. Merger dates in the sample correspond to the period 1980–1998. The standard errors are clustered at firm-year level in columns 1–7. ***, **, and * denote significance at 1%, 5%, and 10%, respectively.

	Difference-in-differences specification				$CPatent^{Treat}$	$CPatent^{Ctrl}$	$Patent^{Treat}$
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
After	−0.091*** (0.020)	−0.106*** (0.019)	−0.106*** (0.022)	0.020** (0.010)	1.013* (0.546)	−0.039 (0.029)	1.181 (0.886)
After $\times T$				−0.248*** (0.037)	−2.093*** (0.666)	−0.007 (0.034)	−2.585 (4.878)
Observations	11,794	11,794	11,794	11,794	11,794	11,794	11,794
R ²	0.134	0.359	0.359	0.362	0.586	0.562	0.575
Other controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	No	No	No	No	No	No
Firm fixed effects	No	Yes	Yes	Yes	Yes	Yes	Yes

(b) Table 5: Main difference-in-differences estimates.

6.5 Table 6: Heterogeneity by merger type, ICM intensity, and target industry

Read:

- Panel A shows that the R&D productivity decline is concentrated in unrelated/diversifying mergers.
- The effect is stronger when the acquirer is a multi-segment firm and when its ICM intensity is high.
- The interaction with *Compete*, *Diversification*, *Diversity*, and *Reallocate* shows that more active ICMs are associated with larger post-merger R&D declines.
- Panel B shows that the treatment effect is stronger in high-technology industries and in more competitive industries.
- These patterns are consistent with the agency problem being most severe when R&D is uncertain, novel, and strategically important.

Economic message: Table 6 is the core mechanism table. It shows that the decline is not a generic merger effect; it is strongest precisely where the internal-capital-market theory predicts weaker incentives for novel R&D.

Table 6

Heterogeneity in treatment effect.

This table reports the results from the quasi-experiment. A difference-in-difference-in-differences specification is employed with R&D productivity measures defined in Appendix A as the dependent variables. The unit of observation is a firm-year. *After* is a dummy variable that takes a value one after the merger event and zero otherwise. *T* is an indicator variable that takes a value one for the treatment group (successful mergers) and zero for the control group (unsuccessful mergers). *Reallocate*^{ICM}, *Diversity*^{ICM}, *Compete*^{ICM} measure the ICM intensity for the bidder averaged over years before the event date. *Dummy*^{ICM} and *Dummy*^{ICM} take a value one if the potential acquirer is in the same SIC as the target and when the potential acquirer is a multi-segment firm, respectively. *Dummy*^{ICM} (used in Panel B) is the sum of *Dummy*^{ICM} and *Dummy*^{ICM}. Merger dates in the sample correspond to the period 1980–1998. The standard errors are clustered at firm-year level. ***, **, and * denote significance at 1%, 5%, and 10%, respectively.

Panel A: Heterogeneity with merger type and ICM intensity of acquirer

	CPatent		Patent		CPatent		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
After	0.021** (0.010)	0.027*** (0.011)	1.081* (0.520)	0.673 (0.417)	0.861 (0.615)	0.151 (0.236)	0.234 (0.135)
After <i>st</i>				-0.323** (0.192)	-0.190* (0.110)	-0.450** (0.228)	-0.406*** (0.127)
After <i>st</i> <i>ICM</i> ^{overloaded}	-0.251*** (0.046)	-0.180*** (0.056)	-1.849*** (0.671)				
After <i>st</i> <i>ICM</i> (1 - <i>Diversity</i> ^{ICM})	-0.025 (0.015)	-0.012 (0.021)	-0.544 (0.388)				
After <i>st</i> <i>ICM</i> <i>Diversity</i> ^{ICM}		-0.149** (0.058)	-1.698* (0.953)				
After <i>st</i> <i>ICM</i> (1 - <i>Diversity</i> ^{ICM})		-0.019 (0.016)	-0.634 (0.413)				
After <i>st</i> <i>ICM</i> <i>Compete</i> ^{ICM}				-0.367** (0.132)			
After <i>st</i> <i>ICM</i> <i>Diversification index</i> ^{ICM}					-0.066* (0.037)		
After <i>st</i> <i>ICM</i> <i>Diversity</i> ^{ICM}						-0.609* (0.359)	
After <i>st</i> <i>ICM</i> <i>Reallocate</i> ^{ICM}							-2.810** (1.223)
Observations	11,794	11,794	11,794	2,299	3,091	3,091	3,091
R ²	0.362	0.463	0.486	0.291	0.307	0.307	0.308
Other controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	No	No	No	No	No	No
Firm fixed effects	No	Yes	Yes	Yes	Yes	Yes	Yes

(a) Table 6, Panel A: Heterogeneity by merger type and acquirer ICM intensity.

Panel B: Heterogeneity with industry characteristics of target

	CPatent							
	Innovativeness				Competitiveness			
	High		Low		High		Low	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
After	0.052 (0.057)	0.052 (0.057)	0.016 (0.034)	0.017 (0.015)	0.097 (0.068)	0.097 (.068)	0.074*** (0.033)	0.078*** (0.021)
After <i>st</i>	-0.289*** (0.067)		-0.104*** (0.033)		-0.344*** (0.065)		-0.201*** (0.052)	
After <i>st</i> <i>ICM</i> (1 – <i>Diversity</i> ^{ICM})		-0.289*** (0.044)		-0.110** (0.054)		-0.345*** (0.045)		-0.202*** (0.066)
After <i>st</i> <i>ICM</i> (1 – <i>Diversity</i> ^{ICM}) <i>Compete</i> ^{ICM}		0.047 (0.199)		-0.011 (0.015)		0.009 (0.010)		-0.018 (0.019)
Observations	3,115	3,115	8,679	8,679	5,897	5,897	5,897	5,897
R ²	0.341	0.341	0.367	0.367	0.355	0.355	0.365	0.365
Other controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes

(b) Table 6, Panel B: Heterogeneity by target innovativeness and competition.

6.6 Table 7 and Table 8: Inventor mechanism and mitigating responses

Read:

- Table 7 separates the extensive and intensive margins of inventor behavior.
- Inventor presence falls after the event, but the decline is similar across treatment and control groups; inventor exit alone cannot explain the treatment effect.
- The average *CPatent* produced by continuing inventors declines in the treatment group, especially in unrelated or high-ICM acquisitions.
- Table 8 examines mitigating forces. Market value, ROA, cash, and R&D intensity do not show a large compensating pattern that explains away the R&D decline.
- The probability that the acquirer enters strategic alliances or joint ventures rises after treatment mergers, and these alliance/JV cases account for a meaningful share of the R&D productivity drop.

Economic message: The mechanism is not simply that creative inventors leave. Instead, inventors who stay become less productive, and acquirers respond by shifting some innovative activity outside the firm boundary.

Table 7

Investor-level evidence.

This table reports the results from the quasi-experiment. A difference-in-difference-in-differences specification is employed to examine whether the inventor stays in the firm after the merger event and the change in R&D productivity of the inventors who stay inside the firm after the merger. The panel data on all the inventors are constructed by partitioning (blocking) the data set into groups and then performing exhaustive comparisons within those groups to construct a list of inventors responsible for the invention. For forming a block, information on the following categories is used: (a) name information (inventor's name), (b) assignee information (the organization to which the patent was granted), (c) location information (variables extracted from the address provided by the individual inventor), (d) class information (patent's technology class), and (e) coauthor information (the list of coauthors on the patent). After is a dummy variable that takes a value one after the merger event and zero otherwise. It is an indicator variable that takes a value one for the treatment group (successful mergers) and zero for the control group (unsuccessful mergers). The unit of observation is at investor-level. Present takes two values for each inventor: a value one (zero) before the event if the inventor is present (not-present) in any of the years before the event and a value of 1 (0) after the event if the inventor is present (not present) in any of the years after the event. Avg-CPatent takes two values for each inventor: the value of average citations per patent for the patents produced by the inventor before the event and the average value of citations per patent the inventor produces after the event. Dummy^{JV} is the sum of Dummy^{successful} and Dummy^{unsuccessful}. Though there are 36,265 observations in the sample in the first three columns, several observations drop out in high regressions with fixed effects. Merger dates in the sample correspond to the period 1980 to 1998. The standard errors are clustered at the investor-level. ***, **, *, and * denote significance at 1%, 5%, and 10%, respectively.

	Investor-level evidence						
	Pr(Present = 1)			Avg-CPatent			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
After	-0.810*** (0.051)	-0.812*** (0.043)	-0.810*** (0.045)	-0.042*** (0.004)	-0.054*** (0.003)	-0.042*** (0.008)	-0.041** (0.018)
After <i>st</i>	0.093 (0.071)	0.093 (0.074)		-0.234** (0.110)	-0.262*** (0.081)		
After <i>st</i> * Dummy st			-0.0141 (0.082)			-0.203** (0.080)	
After <i>st</i> * (1 - Dummy st)			0.3021 (0.073)			0.307 (0.364)	
After <i>st</i> * Compete st							-0.312** (0.152)
Observations	17,708	11,700	11,700	6,480	6,480	6,480	2,071
R ²				0.028	0.212	0.254	0.202
Other controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Inventor fixed effects	No	Yes	Yes	No	Yes	Yes	Yes

(a) Table 7: Investor-level evidence.

Table 8

Mitigating forces.

This table reports the results from the quasi-experiment. A difference-in-difference-in-differences specification is employed with measures of total market-to-book value (Q^{total}), operating profits (ROA^{total}), cash ($Cash/Assets^{total}$), and R&D-to-sales ratio ($R\&D/Sales^{total}$) as the dependent variables in columns 1-5, probability whether a joint venture or strategic alliance was initiated by the acquirer in columns 6-8, and adjusted citations per patent in column 9. The unit of observation is a firm-year. After is a dummy variable that takes a value one after the merger event and zero otherwise. It is an indicator variable that takes a value one for the treatment group (successful mergers) and zero for the control group (unsuccessful mergers). Dummy^{JV} is the sum of Dummy^{successful} and Dummy^{unsuccessful}. JV-Alliance is an indicator variable that takes a value one if the acquirer engages in a new JV or strategic alliance in that year and takes a value zero otherwise. JV-Alliancest is similar to JV-Alliance but takes a value one only for R&D transactions. Merger dates in the sample correspond to the period 1980-1998. The standard errors are clustered at firm-year level. ***, **, and * denote significance at 1%, 5%, and 10%, respectively.

	Treatment effect								
	Q^{total}		ROA^{total}		$Cash/Assets^{total}$		$R\&D/Sales^{total}$		CPatent
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
After	0.171 (0.155)	0.171 (0.167)	-0.019 (0.021)	-0.012 (0.020)	0.006 (0.006)	0.012*** (0.003)	0.006*** (0.001)	0.004** (0.002)	0.025** (0.014)
After <i>st</i>	0.138 (0.157)					0.183*** (0.021)			
After <i>st</i> * Dummy st		0.130 (0.165)	0.008 (0.021)	-0.017 (0.038)	0.005 (0.006)	0.196*** (0.002)	0.220*** (0.003)		-0.131*** (0.027)
After <i>st</i> * (1 - Dummy st)		0.134 (0.154)	0.001 (0.046)	0.007 (0.008)	0.002 (0.003)	-0.006*** (0.001)	-0.004** (0.002)		-0.011 (0.021)
After <i>st</i> * Dummy st * JV-Alliance									-0.095*** (0.036)
Observations	9,951	9,951	10,369	10,369	10,369	8,094	8,094	8,094	8,274
R ²	0.636	0.636	0.616	0.747	0.793				0.483
Other controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

alliance: JV is a variable in which a new locally distinct firm Robinson 2008) which might be contribution in part to a

(b) Table 8: Mitigating forces and movement toward alliances/JVs.

6.7 Table 9: R&D productivity and decentralized R&D budgets

Read:

- Table 9 uses IRI survey data on R&D organization and budget control.
- Panel A shows that the IRI sample contains large, R&D-intensive multi-segment firms.
- Panel B shows that greater headquarters control of R&D budgets is associated with lower $CPatent$.
- The negative relation is concentrated in firms with divisional R&D structures, where the commitment problem is most relevant.
- The Heckman correction in column 3 suggests that selection into the IRI sample matters, but the main decentralization result remains.

Economic message: Decentralization is a governance response to the dark side of internal capital markets. When divisions control more of their own R&D budgets, incentives to pursue novel research appear stronger.

Table 9
R&D Productivity and decentralization of R&D budgets.
This table reports the results that relate citations per patent produced in a multi-segment firm to the degree of decentralization of its R&D budgets. Panel A presents descriptive statistics of some key variables in the IRI subsample. Panel B presents the results of regressions with R&D productivity measures defined in Appendix A as the dependent variable. The unit of observation is a firm-year. $\%H_b\text{Budget}$ measures the proportion of R&D budget of a division that is contributed to by the headquarters. *Divisional* is an indicator variable that takes a value one if the R&D structure of the conglomerate is divisional and zero otherwise. Control variables included in the estimation but unreported for brevity are Sales, RD, Leverage, $EBIDTA/Assets$, $Capex/Assets$, $Cash/Assets$, $NPPE/Assets$, H/I , H/P , and Q . ^a Signifies that data only consider firms that produce at least one patent during a given year. Data are for the period 1980–1998. ^{***}, ^{**}, and ^{*} denote significance at 1%, 5%, and 10%, respectively.

Panel A: Multi-segment firms in IRI sample			
	Mean (1)	Max (2)	Min (3)
Sales (\$ million)	13,315	23,110	91
RD_{it}^a (%)	5.3	21.5	0
Sales Patent ^a	41	982	0
CPatent ^a	0.82	3.51	0
Panel B: CPatent and decentralization of R&D budgets			
	CPatent		
	(1)	(2)	(3)
$\%H_b\text{Budget}$	-1.60 (0.288) ^{***}	0.140 (0.150)	0.138 (0.145)
$\%H_b\text{Budget} \times \text{Divisional}$		-1.69 (0.51) ^{***}	-1.54 (0.52) ^{***}
λ			-0.197 (0.043) ^{***}
Observations	817	817	817
R^2 (%)	0.45	0.45	0.47
Other controls	Yes	Yes	Yes
Time fixed effects	Yes	Yes	Yes

Table 9: R&D productivity and decentralization of R&D budgets.

7 Short summary

- The paper studies whether conglomerate boundaries affect innovative activity.
- The theory is that active internal capital markets can reduce incentives for novel R&D because headquarters may reallocate resources after divisions reveal information.
- Aggregate data show a negative relation between ICM intensity and citations per patent in multi-segment firms.
- The main design compares successful merger targets with failed merger targets.
- Successful targets experience a large decline in patents and citations per patent after acquisition relative to failed targets.
- The decline is concentrated in diversifying/conglomerating mergers and is strongest when the acquirer has an active ICM.
- Inventor-level results show that continuing inventors become less productive; inventor exit alone is not the main channel.
- Acquirers partly respond by using strategic alliances and joint ventures, moving some innovative activity outside firm boundaries.
- Separate IRI evidence shows that decentralized R&D budgets are associated with more novel R&D inside conglomerates.
- The paper’s main contribution is to show that firm boundaries and internal capital allocation shape the type and productivity of innovation.

8 Conclusion

8.1 Core conclusion

Seru shows that conglomerate organizational form can reduce the scale and novelty of R&D produced inside acquired firms. The evidence is strongest for diversifying acquisitions by conglomerates with active internal capital markets. The central result is not merely that conglomerates innovate differently, but that moving a firm inside a conglomerate can causally change its innovation path.

8.2 Economic intuition

Novel R&D depends heavily on soft information. Researchers and divisional managers often understand uncertain projects better than headquarters does. If headquarters can reallocate resources after seeing interim signals, divisions may avoid ambitious projects, hide bad news, or reduce effort in generating novel ideas. This weakens the incentive to pursue research that is uncertain but potentially valuable. The problem is most severe in active ICMs, where reallocation is not a remote possibility but a core feature of the organization.

8.3 Incremental contribution revisited

The paper advances the firm-boundaries literature in three ways. First, it provides causal evidence using failed mergers rather than relying only on cross-sectional comparisons. Second, it measures innovation output and novelty directly using patent data, instead of relying only on investment or firm value. Third, it documents mechanisms through inventor-level productivity, alliance formation, and R&D budget decentralization. The result is a more complete picture of how internal capital allocation affects knowledge production.

8.4 Implications for conglomerates and innovation policy

The findings imply that active headquarters control can be costly when project quality is hard to verify. Conglomerates may still add value through development, commercialization, capital allocation, or resilience to shocks, but they may be poorly suited for novel early-stage research unless they design credible protections for divisional R&D incentives. Delegation, decentralized budgets, and external alliances can help preserve innovative effort.

8.5 Limitations

The evidence applies most directly to publicly traded merger targets with patentable innovation during 1980–1998. Patents and citations are imperfect measures of research quality, especially across industries. The failed-merger design is strong but still depends on the assumption that deal failure is unrelated to unobserved future R&D productivity. The paper also does not claim that conglomeration always destroys value, since acquirers may compensate through other channels.

8.6 Takeaway

The enduring takeaway is that firm boundaries are not neutral. The same internal capital market that may allocate resources efficiently across mature projects can weaken incentives for novel research. For innovation-intensive firms, the design of authority, budget control, and organizational boundaries can shape what kind of ideas are produced inside the firm.